

BACnet/SC Hub (B-SCHUB) - User Manual

Version 1.5.0 · Chipkin Automation Systems · <https://github.com/chipkin/BACnetProfileExample-B-SCHUB-CPP>

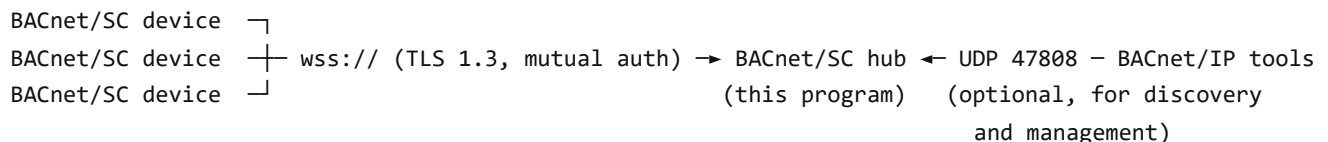
This manual is for people who install, configure and run the hub. `BACnetExampleBSCHUB --version` prints the version you are running, with the CAS BACnet Stack and `common/` helper versions. For how the code works and how to build your own product from it, see [TUTORIAL.md](#).

Contents

1. [What the hub does](#)
2. [Installation](#)
3. [Quick start](#)
4. [Certificates](#)
5. [Configuration](#)
6. [Connecting devices](#)
7. [Managing certificates over BACnet](#)
8. [Status page and HTTP endpoints](#)
9. [Security](#)
10. [Troubleshooting](#)
11. [The BACnet device](#)
12. [Version, licensing and support](#)

1. What the hub does

BACnet Secure Connect (BACnet/SC, ANSI/ASHRAE 135 Annex AB) carries BACnet over TLS-secured WebSockets instead of plain UDP. Devices on a BACnet/SC network don't talk to each other directly: each one connects to a **hub**, which relays their traffic - the role a BACnet/IP broadcast domain plays for UDP devices. This program is that hub, built on the [CAS BACnet Stack](#).



It implements the **B-SCHUB** device profile (ANSI/ASHRAE 135 Annex L):

- **BACnet/SC hub function** (BIBB NM-SCH-B) on Network Port 2: listens on `wss://0.0.0.0:4443/` (TLS 1.3, mutual certificate authentication, WebSocket subprotocol `hub.bsc.bacnet.org`) and relays traffic between the connected devices (up to 4 at a time - see [Connection limit](#)), including this hub's own BACnet device.
- **Optional hub connector**: with `--sc-hub-uri`, the hub also connects out to another BACnet/SC hub (with an optional failover hub).
- **BACnet/IP** on Network Port 1 (UDP 47808), on by default: discovery (Who-Is/I-Am, Who-Has/I-Have), ReadProperty, ReadPropertyMultiple and DeviceCommunicationControl. It can be turned off for a BACnet/SC-only site ([BACnet/SC only](#)).
- **Certificate management over BACnet**: a client can add an issuer and replace the hub's operational certificate (clause 19.8.3).

- **Status page, health and metrics** over HTTP or HTTPS, for people and for monitoring tools.
- **Built-in lab certificate generator**, including a ready-to-import connection file for each device.

This build is for **evaluation and testing**: it accepts at most 4 BACnet/SC devices at a time. For a production hub, contact support@chipkin.com.

Supported BIBBs

BIBB	Name
DS-RP-B	Data Sharing - ReadProperty - B
DS-RPM-B	Data Sharing - ReadPropertyMultiple - B
DM-DDB-B	Device Management - Dynamic Device Binding - B
DM-DOB-B	Device Management - Dynamic Object Binding - B
DM-DCC-B	Device Management - DeviceCommunicationControl - B
NM-SCH-B	Network Management - BACnet/SC Hub Function - B

Services executed

Service	Use
ReadProperty, ReadPropertyMultiple	Read any property of any object.
Who-Is / I-Am, Who-Has / I-Have	Discovery. The hub also broadcasts I-Am at startup over BACnet/IP.
DeviceCommunicationControl	Silence or resume the device, optionally with a password.
AtomicReadFile	Read the certificate File objects.
AtomicWriteFile, WriteProperty	Certificate management only: write a certificate into File 1, 3 or 4, and set their File_Size. No other property is writable.
ReinitializeDevice	ACTIVATE_CHANGES or WARMSTART applies written certificates. Other states are refused.

The [PICS](#) lists every object and property and who answers it.

2. Installation

Download from the [Releases page](#):

Download	What it is
BACnetSCHub-<version>-setup.exe	Windows installer. Installs the program in C:\Program Files\Chipkin\BACnet SC Hub; settings, certificates and logs go in C:\ProgramData\Chipkin\BACnetSCHub. Optionally sets up the BACnetSCHub Windows service (starts on boot, restarts on failure) and firewall rules for UDP 47808 and TCP 4443.

Download	What it is
bacnet-schub-hub_<version>_amd64.deb	Debian/Ubuntu package: <code>sudo apt install ./bacnet-schub-hub_<version>_amd64.deb</code> . The program goes in <code>/opt/bacnet-schub</code> , settings in <code>/etc/bacnet-schub/hub.conf</code> , certificates in <code>/etc/bacnet-schub/certs</code> , logs in <code>/var/log/bacnet-schub</code> , and the bacnet-schub-hub systemd service is enabled and started, running as the <code>bacnethub</code> user.
BACnetSCHub-<version>-linux-x64.tar.gz	The same for other Linux distributions: unpack it and run <code>sudo ./install.sh</code> . <code>sudo ./uninstall.sh</code> removes it (<code>--purge</code> also removes settings and certificates).
BACnetExampleBSCHUB.exe, BACnetExampleBSCHUB	The bare program, to run from a folder of your choice.
manual.pdf, fact-sheet.pdf, PICS.pdf	This manual, the two-page fact sheet, and the PICS.
sbom-windows.cdx.json, sbom-linux.cdx.json	Software bill of materials: the exact CAS BACnet Stack, OpenSSL, libwebsockets and other library versions.
SHA256SUMS.txt	SHA-256 checksums of every download: <code>sha256sum -c SHA256SUMS.txt --ignore-missing</code> .

Both installers make a lab certificate set if the certificate folder is empty, so the hub starts listening straight away; replace it for production (see [Certificates](#)). Settings and certificates are kept when you upgrade or uninstall.

Signed downloads. The Windows program and the Windows installer are code-signed by Chipkin (Azure Artifact Signing, SHA-256 with a timestamp); check them with `Get-AuthenticodeSignature <file>`. Every download - the Linux archive and .deb included - also has a signed build-provenance attestation: `gh attestation verify <file> -repo chipkin/BACnetProfileExample-B-SCHUB-CPP` confirms it was built by this project's release workflow. `SHA256SUMS.txt` lists every file's checksum. See [code-signing.md](#).

Supported platforms

Windows 10/11 and Windows Server 2016 or later (x64). Linux x64: the release build is made on the current Ubuntu LTS, so it needs a distribution with that glibc version or newer; for an older distribution, build from source (see the README). The hub also builds from source on macOS.

The hub runs fine on Windows 10, but BACnet/SC tools on a Windows 10 computer that use Windows' own TLS (such as YABE) can't connect to it or to any other BACnet/SC hub: BACnet/SC requires TLS 1.3, and Windows 10 can't make TLS 1.3 client connections. The hub logs a warning about this when it starts on Windows 10 ([#39](#)).

Firewall

Allow in:

Port	Protocol	For
4443	TCP	BACnet/SC devices connecting to the hub (<code>--sc-port</code>).
47808	UDP	BACnet/IP discovery and management (<code>--port</code>), unless BACnet/IP is off.
8080	TCP	The status page and HTTP endpoints (<code>--http-port</code>) - only if you bind it beyond 127.0.0.1.

The Windows installer can add the first two rules for you.

Running as a service

Windows. The installer sets up the **BACnetSCHub** service. To do it by hand, from an Administrator command prompt:

```
BACnetExampleBSCHUB --install-service --config C:\ProgramData\Chipkin\BACnetSCHub\hub.conf
sc start BACnetSCHub
BACnetExampleBSCHUB --uninstall-service
```

The service runs `BACnetExampleBSCHUB --service --config <file>` with the config file's folder as its working directory, so relative paths in it (`sc-cert-dir = certs`, `log-file = logs\hub.log`) are relative to that folder. There is no console, so use `log-file` to keep a log. The service starts on boot and is restarted after 5, 30 and then 60 seconds if it fails.

Linux (systemd). The `.deb` and `install.sh` install `bacnet-schub-hub.service`, which runs the hub with `/etc/bacnet-schub/hub.conf`, restarts it on failure, and sends its output to the journal as well as `log-file`:

```
systemctl status bacnet-schub-hub
journalctl -u bacnet-schub-hub -f
sudo systemctl restart bacnet-schub-hub      # after editing hub.conf
```

`systemctl stop` sends `SIGTERM`, which stops the hub cleanly, like pressing `q`.

3. Quick start

From a folder where you unpacked the program:

```
# 1. Make a lab certificate set: a CA, the hub's certificate, and 3 device
#   certificates (clients/client-01 .. client-03).
BACnetExampleBSCHUB --generate-certs
```

```
# 2. Start the hub.
BACnetExampleBSCHUB
```

1. Open <http://127.0.0.1:8080/> in a browser. The status page shows the version, whether the BACnet/SC hub is listening, and the connected devices.
2. Give each BACnet/SC device one `certs/clients/<label>/` folder. In the [CAS BACnet Explorer](#), import that folder's `bacnetsc.config` (see [Connecting devices](#)).

On Windows the program is `BACnetExampleBSCHUB.exe`. A typical start-up:

```
BACnet B-SCHUB (BACnet/SC Hub) Example - C++ v1.5.0
CAS BACnet Stack version: 6.0.23.0
Common helper (common/) version: 3.0.0
FYI: Listening for BACnet/IP on UDP port 47808 (Network Port 1).
FYI: Device 389022 ("Chipkin Example B-SCHUB") ready. Vendor ID 389. Press 'h' for help, 'm' for a
health/metrics snapshot.
BACnet/SC: listening for WebSocket/TLS connections on wss://0.0.0.0:4443/ (subprotocol
"hub.bsc.bacnet.org", TLS 1.3, mutual auth)
```

At start-up the hub also checks its certificates and logs each one's subject, days until expiry (a warning under 30 days), whether the private key matches, and the subjectAltName entries.

Console keys

Key	Action
h	Help and version.
m	Health and metrics snapshot.
up / down	Change Analog Input 1 by 1.1.
q	Quit.

4. Certificates

BACnet/SC runs over TLS with **mutual authentication**: the hub and every device present a certificate, and each side accepts the other only if that certificate was signed by an issuer (CA) it trusts.

Lab certificates

The hub can make a complete lab certificate set itself:

```
BACnetExampleBSCHUB --generate-certs      # CA + hub + 3 device certificates
BACnetExampleBSCHUB --generate-certs 10   # ... or any number of devices
```

This writes PEM files to --sc-cert-dir (default ./certs) and exits. The file names follow the Network Port properties that carry them (ANSI/ASHRAE 135 clause 12.56):

File	What it is
operational-certificate.pem	The hub's certificate (File 1, Operational_Certificate_File). Its subjectAltName lists localhost, 127.0.0.1, this computer's host name and the hub URI's host.
private-key.pem	The hub's private key. Private .
certificate-signing-request.pem	CSR for the hub's key (File 2, Certificate_Signing_Request_File).
issuer-certificate.pem	The lab CA (File 3, Issuer_Certificate_Files). Every device needs a copy.
issuer-private-key.pem	The CA's private key, only used to sign more devices (--add-client-certs, --sign-csr). Private . Keep it off the network.
clients/<label>/	One folder per device: its operational-certificate.pem, private-key.pem (private), issuer-certificate.pem, bacnetsc.config (CAS BACnet Explorer), and for Windows tools such as YABE <label>.pfx (certificate + key + issuer, empty password, private), issuer-certificate.cer (DER) and yabe-bacnetsc.config.
certificates.txt	Every certificate's label, location, serial number, expiry and SHA-256 fingerprint.
readme.txt	A walkthrough of the folder, including which files are private.

The hub adds two files of its own: `issuer-certificate-2.pem` (File 4, once a second issuer is written over BACnet) and `trusted-issuers.pem` (every issuer from both slots; this is what TLS trusts). You can add a third, `issuer-cr1.pem`, to revoke device certificates (see [Security](#)).

Device certificates are labeled by folder and by subject Common Name (Chipkin Example B-SCHUB client-01), so the hub's log shows which device connected.

Adding devices later. Sign more device certificates with the *existing* CA. Numbering continues, and a running hub trusts them straight away:

```
BACnetExampleBSCHUB --add-client-certs 2 # clients/client-04, client-05
BACnetExampleBSCHUB --add-client-certs 3 --cert-label ahu # clients/ahu-01 .. ahu-03
```

Signing a device's own CSR. When a device makes its own key and sends you a certificate signing request (PEM or DER), sign it with the existing CA. The private key never leaves the device:

```
BACnetExampleBSCHUB --sign-csr ahu7.csr --cert-label ahu-7 # clients/ahu-7/
```

The hub first checks that the request's signature verifies and that its key is ECDSA P-256 or stronger, or RSA 2048-bit or stronger. `clients/ahu-7/` then gets the device's `operational-certificate.pem`, `issuer-certificate.pem` (and `.cer`) and `bacnetsc.config`, with no private key, `.pfx` or YABE file. Send that folder back to the device's owner. The certificate keeps the request's subject and public key. Everything else comes from the device profile above: 825 days, EKU `clientAuth`, no `subjectAltName`. Any extensions the request asks for are ignored. Without `--cert-label` the folder is the next `client-NN`. A running hub trusts the new certificate straight away.

To make a key and a CSR for a device with this program, for example on the device owner's own computer:

```
BACnetExampleBSCHUB --generate-csr --cert-label ahu-7 # clients/ahu-7/private-key.pem +
certificate-signing-request.pem
```

`--generate-csr` needs no CA. If you sign the request in the same folder (`--sign-csr clients/ahu-7/certificate-signing-request.pem`, where the label comes from the folder), the certificate lands next to its key, and the folder also gets the `.pfx` and YABE files.

Starting over. `--generate-certs` won't replace an existing CA, because every certificate already handed out would stop working. Add `--force` to delete the whole set, including `clients/`, and make a new one.

The lab profile is ECDSA P-256 with SHA-256; the CA is valid for 10 years and device certificates for 825 days. Folders made by earlier releases (`hub.crt`, `hub.key`, `ca.crt`) still work.

Production certificates

For a production site, use your own CA: have it sign the hub's CSR (File 2), and install the result on disk or [over BACnet. production-certificates.md](#) walks through it: using a site or enterprise CA (root and intermediate), getting the hub's certificate signed, rotating certificates and changing the CA without an outage, revocation, what the certificates should contain, and protecting private keys.

Password-protected private keys

A certificate package may hold an encrypted private key (the file starts `-----BEGIN ENCRYPTED PRIVATE KEY-----`). The hub asks for its password **once**, when it starts, with the typing hidden, and checks it by decrypting the key; a wrong password is asked for again, up to three times. It keeps the password in memory and uses it everywhere the

key is needed - the listener, the connector's every connect attempt, the HTTPS status page - so it is never asked for again during the run.

A service has no console to ask on: set the password in the config file as `sc-key-password` (config file only, like `dcc-password`; restrict the file's permissions as described in [Configuration file](#)). When it is set, the hub doesn't ask.

5. Configuration

BACnetExampleBSCHUB [options]

Options

Option	Default	Meaning
<code>--port <n></code>	47808	BACnet/IP UDP port.
<code>--bacnet-ip <on/off></code>	on	off runs BACnet/SC only - see BACnet/SC only .
<code>--deviceID <n></code>	389022	BACnet device instance.
<code>--device-name <name></code>	Chipkin Example B-SCHUB	The Device's Object_Name. Must be unique on the BACnet internetwork, so name each hub. 1 to 128 bytes of UTF-8; for a non-ASCII name use device-name in a config file saved as UTF-8.
<code>--ip-network-number <n>, - --sc-network-number <n></code>	not set	Network_Number of Network Port 1 / 2 (1..65534), reported with quality configured. Unset ports report 0, quality unknown.
<code>--log-file <path></code>	none	Also write everything the hub prints to this file (see Logging).
<code>--log-max-size-mb <n>, -- log-max-files <n></code>	10, 5	Rotate the log file at this size, keeping this many old files (<path>. <i>1</i> , <i>2</i> , ...).
<code>--sc-port <n></code>	4443	BACnet/SC hub (wss://) port.
<code>--sc-cert-dir <dir></code>	./certs	Certificate folder.
<code>--sc-hub-uri <wss://host:port/></code>	off	Also connect out to this BACnet/SC hub.
<code>--sc-failover-uri <wss://host:port/></code>	off	Failover hub for --sc-hub-uri.
<code>--sc-max-hub-connections <n></code>	4	Maximum BACnet/SC devices connected at once, 1 to 4 (see Connection limit).
<code>--sc-rate-limit <n></code>	10	Maximum new BACnet/SC connection attempts per second from any one source address (0 = no limit). Excess attempts are refused before the TLS handshake.
<code>--sc-rate-limit-total <n></code>	50	Maximum new BACnet/SC connection attempts per second for the whole listener (0 = no limit).
<code>--sc-accept-hub-without- hello</code>	off	Compatibility: let --sc-hub-uri connect to a hub that omits the Hello option. See BACnet/SC compatibility .
<code>--sc-accept-device- without-hello</code>	off	Compatibility: accept a device whose Connect-Request omits the Hello option, such as YABE. See BACnet/SC compatibility .

Option	Default	Meaning
<code>--http-port <n></code>	8080	Status page and HTTP endpoints.
<code>--http-bind <addr></code>	127.0.0.1	Interface for the HTTP listener. See Security before changing it.
<code>--http-tls</code>	off	Serve the HTTP endpoints over HTTPS.
<code>--http-tls-cert <file>, --http-tls-key <file></code>	the hub's operational certificate and key	Certificate and key for <code>--http-tls</code> .
<code>--sc-keylog-file <file></code>	off	Debugging only. Append each BACnet/SC connection's TLS session secrets to <file>, so Wireshark can decrypt a capture. Command line only. See Decoding BACnet/SC traffic in Wireshark .
<code>--config <path></code>	none	Read settings from a config file (below).
<code>--generate-certs [n]</code>	3	Make a lab certificate set with n device certificates, then exit.
<code>--add-client-certs [n]</code>	1	Sign n more device certificates with the existing CA, then exit.
<code>--generate-csr</code>	-	Make a private key and a certificate signing request for one device in <code>clients/<label>/</code> , then exit.
<code>--sign-csr <file></code>	-	Sign a device's certificate signing request with the existing CA into <code>clients/<label>/</code> , then exit.
<code>--cert-label <prefix></code>	client	Label for new device certificates. With <code>--generate-csr</code> or <code>--sign-csr</code> it's the folder name as given (default: the next <code>client-NN</code>).
<code>--cert-hub-uri <wss://host:port/></code>	this computer's IPv4 and <code>--sc-port</code>	Hub URI written into each device's <code>bacnetsc.config</code> (and the hub certificate's <code>subjectAltName</code>).
<code>--force</code>	-	With <code>--generate-certs</code> : replace an existing certificate set.
<code>--install-service, --uninstall-service</code>	-	Windows: set up or remove the BACnetSCHub service (see Running as a service).
<code>--help, --version</code>	-	Usage, or version information.

An unknown option, a stray argument or an option missing its value is an error. The on/off switches (`--http-tls`, `--sc-accept-hub-without-hello`, `--sc-accept-device-without-hello`) mean "on" alone and also take on or off, so the command line can turn off what the config file turned on.

Configuration file

`--config <path>` reads `key = value` lines. A line starting with `#` is a comment; a `#` later in a line is part of the value, so passwords and tokens may contain one. `example.conf` (shipped with the program) lists every key with its default: `device-id`, `device-name`, `ip-network-number`, `sc-network-number`, `log-file`, `log-max-size-mb`, `log-max-files`, `port`, `bacnet-ip`, `sc-port`, `sc-cert-dir`, `sc-hub-uri`, `sc-failover-uri`, `dcc-password`, `sc-key-password`, `http-upload-token`, `http-port`, `http-bind`, `http-tls`, `http-tls-cert`, `http-tls-key`, `sc-max-hub-connections`, `sc-rate-limit`, `sc-rate-limit-total`, `sc-accept-hub-without-hello` and `sc-accept-device-without-hello`. On/off keys take

true/false (or on/off, yes/no, 1/0). A command-line option always wins over the config file. An unknown key or bad value is logged and skipped.

dcc-password, sc-key-password and http-upload-token can only be set in the config file, so they never show up in process listings or shell history. `sc-key-password` is the password of an encrypted private key (see [Password-protected private keys](#)). `dcc-password` protects DeviceCommunicationControl and ReinitializeDevice; `http-upload-token` is the separate secret for the HTTP certificate upload (use a different value - the hub warns if they match). Both are compared in constant time. Restrict the file's permissions (`chmod 600 hub.conf`, or on Windows `icacls hub.conf /inheritance:r /grant:r *S-1-5-18:F *S-1-5-32-544:F "%USERNAME%:F"`, which keeps access for SYSTEM and Administrators; if the installer set up the service, also add "NT SERVICE\BACnetSCHub:R", the account it runs as); the hub warns at start-up if the file looks readable by other users. The Windows installer already restricts `C:\ProgramData\Chipkin\BACnetSCHub` to SYSTEM, Administrators and the service's own account, NT SERVICE\BACnetSCHub.

Logging

Everything the hub prints goes to the console. With `--log-file <path>` (config: `log-file`) it also goes to that file, which is appended to across restarts. When the file reaches `--log-max-size-mb` (default 10 MB) it is renamed `<path>.1` (the previous `.1` becomes `.2`, and so on) and a new file is started; `--log-max-files` (default 5) old files are kept. Every line written through the hub's logger starts with a UTC timestamp and a level ([INFO], [WARN], [ERROR]). Connection audit lines start with `SC audit:.`

BACnet/SC only

Some sites want no unencrypted BACnet traffic at all. `--bacnet-ip off` (config: `bacnet-ip = off`) turns BACnet/IP off: the hub opens no UDP port, Network Port 1 doesn't exist (it isn't in the Device's `Object_List`), and the device is reachable only over BACnet/SC - discovery (Who-Is/I-Am) and every other service go through the hub. It doesn't send an I-Am at start-up, since no BACnet/SC device is connected yet; devices find it with Who-Is once they connect.

The hub warns at start-up if BACnet/IP is off and BACnet/SC isn't listening (missing or bad certificates, say), because the device can't then be reached over BACnet at all. BACnet/SC keeps retrying every 5 seconds, and `/health` reports degraded until it is listening; the status page and `/health (bacnet_ip_enabled)` show which mode the hub is in.

BACnet/SC compatibility

ANSI/ASHRAE 135-2024 (clause AB.2.2) requires the BACnet/SC Connect-Request and Connect-Accept messages to carry a *Hello* option. By default the hub follows the standard strictly and refuses a connection that leaves it out.

`--sc-accept-hub-without-hello` (config: `sc-accept-hub-without-hello = true`) relaxes this for the **hub connector** only: when the hub connects out to another hub (`--sc-hub-uri`), it accepts that hub's Connect-Accept without Hello and treats the other hub as having no optional capabilities. Turn it on only for a hub that needs it; the hub logs a warning at start-up while it is on.

`--sc-accept-device-without-hello` (config: `sc-accept-device-without-hello = true`) does the same for **devices** connecting to this hub: it accepts a Connect-Request without Hello, which YABE sends, and records the device as having no optional capabilities. Without it, such a device is refused and the hub logs `Connect-Request refused` by the hub: "Connect messages require the Hello destination option". Turn it on only while a device like that needs to connect; the hub logs a warning at start-up while it is on.

The two settings are independent; either, both or neither can be on.

Connection limit

This example accepts **at most 4 BACnet/SC devices at a time**. It is for evaluation and testing, not for production. You can lower the limit with `--sc-max-hub-connections` or `sc-max-hub-connections` in the config file, but not raise it: asking for more than 4 prints an error and the hub shuts down.

For a production BACnet/SC hub with more connections, contact Chipkin at support@chipkin.com.

6. Connecting devices

For each BACnet/SC device:

1. Give it its own `certs/clients/<label>/` folder. Don't share a folder between devices: the hub couldn't tell them apart.
2. **CAS BACnet Explorer**: import the folder's `bacnetsc.config`. It names the hub URI and the folder's three PEM files, so keep them together.
3. **YABE (Yet Another BACnet Explorer)**: in *Communication Channel* -> *BACnet/Secure Connect*, **Select** the folder's `yabe-bacnetsc.config`, then **Start**. It points at the folder's `<label>.pfx` (certificate, key and issuer, with an empty password - keep it private) and `issuer-certificate.cer` by absolute path, so re-select it if you move the folder. YABE uses Windows' own TLS, which can't make the TLS 1.3 connections BACnet/SC requires on Windows 10: use Windows 11 / Server 2022 or later ([#39](#)).
4. **Any other BACnet/SC device**: install `operational-certificate.pem` and `private-key.pem` as its operational certificate and key, and `issuer-certificate.pem` as its issuer certificate. Set its primary hub URI to `wss://<hub address>:4443/`.

The hub URI in `bacnetsc.config` is this computer's LAN address unless you generated the set with `--cert-hub-uri`. The hub logs refused TLS handshakes with the reason, and keeps an audit trail of every connection (lines start with `SC audit:`): the device's address and the subject of the certificate it presented, then its BACnet/SC VMAC and device UUID once the hub accepts it, and the same identity again when it disconnects:

```
2026-09-26 18:10:21 [INFO] SC audit: peer "wss://0.0.0.0:4443/|client=3" connected from
10.0.0.31:64150, certificate "O = Chipkin Automation Systems (lab test), CN = Chipkin Example B-SCHUB
client-01"
2026-09-26 18:10:21 [INFO] SC audit: peer "wss://0.0.0.0:4443/|client=3" (10.0.0.31:64150) is BACnet/SC
device VMAC 0a:0b:0c:0d:0e:0f, UUID 50515253-5455-5657-5859-5a5b5c5d5e5f, certificate "O = Chipkin
Automation Systems (lab test), CN = Chipkin Example B-SCHUB client-01" - connected
```

The status page lists the devices connected right now the same way.

7. Managing certificates over BACnet

A BACnet client can change the hub's certificates with the BACnet/SC certificate procedures (ANSI/ASHRAE 135 clause 19.8.3), for example from the [CAS BACnet Explorer](#)'s certificate page:

1. **Write** - `WriteProperty File_Size = 0`, then `AtomicWriteFile` the new PEM certificate: File 1 for the hub's own certificate, File 3 or 4 for an issuer.
2. **Staged** - reading the File back shows the new certificate and Network Port 2's `Changes_Pending` is `TRUE`, but nothing is saved yet and the hub keeps using its current certificates.
3. **Activate** - `ReinitializeDevice ACTIVATE_CHANGES` (or `WARMSTART`). The hub checks the staged certificates first: each must be a PEM certificate, at least one issuer must remain, and the hub certificate must match its

private key and chain to an issuer. If they pass, it saves them and restarts BACnet/SC; devices reconnect under the new certificates. If not, it answers `INVALID_CONFIGURATION_DATA` and nothing changes, so a mistake can't lock the hub out. The staged writes stay, so the client can correct them and activate again.

Supported procedures:

- **Add an issuer** - write the new CA into slot 2 (File 4). Devices signed by either CA are then accepted, which lets a site move to a new CA gradually.
- **Replace the hub certificate** - read the CSR (File 2), have your CA sign it, write the result to File 1, activate.
- **Replace the hub certificate with a new key pair** - write `GENERATE_CSR_FILE` (9) to Network Port 2's Command. The hub makes a new key pair and a new CSR (File 2) with the same subject, saving the new key as `private-key-pending.pem`; it keeps using its current key and certificate meanwhile. Have the new CSR signed, write the result to File 1 and activate: the new key then replaces `private-key.pem`. Refused with `INVALID_VALUE_IN_THIS_STATE` while certificate writes are staged - and, until #41 is fixed, on a freshly started hub, which reads `Changes_Pending TRUE` with nothing staged: send `ReinitializeDevice ACTIVATE_CHANGES` once first. If the hub stops between saving the new certificate and swapping the key, it finishes the swap at the next start.
- **Discard staged writes** - write `DISCARD_CHANGES` (1) to Network Port 2's Command. The staged certificate writes are dropped, the File objects read the files on disk again, and `Changes_Pending` goes back to `FALSE`. A restart also discards staged writes. Until #41 is fixed, don't send it to a freshly started hub: it also reverts the stack's start-up state, after which `Changes_Pending` no longer follows certificate writes (restart the hub to recover).

Known issue: right after the hub starts, Network Port 2 reads `Changes_Pending = TRUE` (and `Current_Health` may report `invalid-configuration-data`) although nothing has been written. The CAS BACnet Stack leaves the hub's own start-up certificate settings pending (`cas-bacnet-stack#2866`, #41). The certificates in use are correct. The first `ReinitializeDevice ACTIVATE_CHANGES` clears it.

If `dcc-password` is set, `ReinitializeDevice` requires it.

8. Status page and HTTP endpoints

The hub serves HTTP on `--http-port` (default 8080), on `127.0.0.1` only by default. With `--http-tls` it serves HTTPS instead (TLS 1.2 or 1.3), using the hub's own `operational-certificate.pem` and `private-key.pem` unless `--http-tls-cert/--http-tls-key` name another certificate - for example one your browsers already trust. With the lab certificates, point curl at the lab CA: `curl --cacert certs/issuer-certificate.pem https://localhost:8080/health` (with Windows' own curl, add `--ssl-no-revoke`: the lab CA publishes no revocation list for Windows to check). The HTTPS certificate is loaded at start-up.

Path	What it returns
GET /	Status page for a browser: versions, device, health, BACnet/SC state, metrics, the connected devices (address, certificate, VMAC and UUID), and links to this project and the CAS BACnet Stack. Refreshes every 5 seconds.
GET /health	<i>Is the hub working?</i> JSON with status ok (HTTP 200) or degraded (HTTP 503 - the BACnet/SC hub isn't listening, usually because of missing or bad certificates). Point uptime monitors and load balancers here.

Path	What it returns
GET /metrics	<i>How much is it doing?</i> JSON counters: uptime, connected devices, connects, disconnects, rate-limit refusals, messages and bytes in and out, and connections closed because the device stopped reading (sc_tx_queue_overflows).
POST /certs/<slot>	Upload a certificate file (operational, csr, issuer1, issuer2). Needs Authorization: Bearer <http-upload-token>; disabled when no http-upload-token is set. At most 5 attempts a minute from one address and 30 in total; more get HTTP 429 and a log line.

```
$ curl -s http://127.0.0.1:8080/health
{"status":"ok","version":"1.5.0","uptime_seconds":154,"bacnet_ip_enabled":true,"sc_hub_function_listeni
ng":true,"sc_hub_connections_current":1,"staged_certificate_changes":false,"tls_keylog":false}
```

```
$ curl -s http://127.0.0.1:8080/metrics
{"uptime_seconds":154,"uptime":"2m
34s","sc_hub_connections_current":1,"sc_hub_connections_max":4,"sc_total_connects":3,"sc_total_disconne
cts":2,"sc_rate_limit_rejections":0,"sc_rx_messages":12,"sc_rx_bytes":456,"sc_tx_messages":12,"sc_tx_by
tes":456,"sc_tx_queue_overflows":0}
```

Uploading a certificate

An upload is checked exactly like a certificate written [over BACnet](#) before anything is saved: it must be a PEM certificate, and the certificates as they would be afterwards must still let the hub run (the hub certificate matches its private key and chains to an issuer, and at least one issuer remains). If they pass, the file is written atomically and BACnet/SC reloads with it straight away; devices reconnect. If not, the hub answers HTTP 400 with the reason and nothing changes. An upload to `csr` must be a certificate signing request for the hub's own private key. While a certificate change made over BACnet is waiting for `ACTIVATE_CHANGES`, uploads are refused with HTTP 409.

```
curl -X POST --data-binary @new-hub-cert.pem \
-H "Authorization: Bearer $UPLOAD_TOKEN" http://127.0.0.1:8080/certs/operational
```

9. Security

- **BACnet/SC** uses TLS 1.3 only, with mutual authentication. A device is accepted if its certificate chains to an issuer in File 3 or 4, and isn't revoked (below).
- **Certificate revocation.** Put your CA's certificate revocation list(s) in `issuer-cr1.pem` in `--sc-cert-dir` (PEM; one CRL per issuer, concatenated). The hub then refuses any device whose certificate is on a list. It checks the file every 5 seconds: a new or changed file restarts BACnet/SC, every device reconnects, and revoked ones are refused, so no restart is needed. Revocation checking fails closed: while the file exists, a device whose issuer has no CRL in it, or whose CRL is past its next-update date, is refused too, and a file with no readable CRL stops BACnet/SC (/health reports degraded). Keep the CRLs current. Without the file, revocation isn't checked. The start-up log lists each CRL, how many certificates it revokes, and when it expires.
- **Host names aren't checked** when the hub connects out to another hub. BACnet/SC certificates identify devices, not DNS names; the certificate chain is still verified.
- **The HTTP listener** binds to 127.0.0.1 by default. If you need it from another machine, turn on `--http-tls` before `--http-bind`, so the upload token and status data are encrypted (or use an SSH tunnel or a TLS reverse proxy). The hub logs a warning on every start bound off loopback over plain HTTP. `/`, `/health` and `/metrics` have no authentication, even over HTTPS.

- **Private keys** (`private-key.pem`, `issuer-private-key.pem`) must stay private. Keep the CA's key off the hub in production.
- **TLS key log.** `--sc-keylog-file` writes the BACnet/SC session secrets to a file, so that anyone with the file and a capture can read the traffic. It's for debugging only, and is off unless given on the command line (see [Decoding BACnet/SC traffic in Wireshark](#)). `/health's` `tls_keylog` tells a monitor when it is on.
- **Rate limiting:** new BACnet/SC connection attempts are limited per source address (`--sc-rate-limit`, default 10 a second) and for the whole listener (`--sc-rate-limit-total`, default 50 a second), before the TLS handshake. One flooding host uses up only its own allowance, so devices on other addresses can still reconnect. The hub logs each refusal with the address. A device that stops reading is disconnected after 64 messages are waiting for it.
- **Unencrypted BACnet.** BACnet/IP (UDP 47808) is plain text. Turn it off with `--bacnet-ip off` where only BACnet/SC should be used.

To report a vulnerability, see [SECURITY.md](#).

10. Troubleshooting

Symptom	Cause and fix
cannot start listening ... certificate file(s) missing/unreadable	No certificates in <code>--sc-cert-dir</code> . Run <code>--generate-certs</code> , or point <code>--sc-cert-dir</code> at your certificates. BACnet/IP keeps working. <code>/health</code> reports degraded.
<code>lws_create_context</code> failed ... retrying every 5 s	The BACnet/SC port is in use, or a certificate file doesn't load. The hub keeps retrying; fix the cause and it starts listening without a restart.
SC TLS handshake REJECTED - client certificate failed verification	The device's certificate isn't signed by an issuer the hub trusts, has expired, or is revoked (certificate revoked, unable to get certificate CRL when <code>issuer-crl.pem</code> has no CRL for its issuer, CRL has expired). Check it with <code>openssl verify -CAfile issuer-certificate.pem operational-certificate.pem</code> (add <code>-crl_check -CRLfile issuer-crl.pem</code> with a CRL).
private key ... DOES NOT MATCH at start-up	<code>private-key.pem</code> and <code>operational-certificate.pem</code> are from different sets.
Devices that connected before an update to 1.5.0 no longer do	The default BACnet/SC port is 4443; before 1.5.0 it was 47819. Either set <code>sc-port = 47819</code> in the config file (or <code>--sc-port 47819</code>), or point the devices (and the firewall rule) at port 4443. The start-up log names the port in use.
A device can't reach the hub	Check the hub URI in its <code>bacnetsc.config</code> , TCP 4443 in the firewall, and whether 4 devices are already connected (the connection limit). The SC audit: lines show what the hub saw.
refusing a WebSocket upgrade ... (HTTP 400)	The device asked for a WebSocket subprotocol other than <code>hub.bsc.bacnet.org</code> . Check its BACnet/SC settings.
Connect-Request refused by the hub	Another connected device has the same VMAC, or the hub is full.
SC rate limit: refusing a new connection	A device (or something else) is reconnecting faster than <code>--sc-rate-limit</code> allows.

Symptom	Cause and fix
ERROR: TOO MANY BACnet/SC CONNECTIONS REQUESTED and the hub exits	sc-max-hub-connections is above 4. Set it to 4 or less. See Connection limit .
/health is degraded and BACnet/IP is off	BACnet/SC isn't listening, so the device is unreachable over BACnet. Fix the certificates, or run with --bacnet-ip on.
Red Error: lines at start-up	Normal CAS BACnet Stack debug output (e.g. the hub hearing its own broadcast I-Am).
A BACnet/SC tool on Windows 10 (e.g. YABE) fails with A call to SSPI failed / a TLS handshake error, and the hub logs nothing	Windows 10's own TLS can't do TLS 1.3, which BACnet/SC requires. The hub warns about this at start-up on Windows 10. Run the tool on Windows 11 / Server 2022 or later, or use a client with its own TLS 1.3 (CAS BACnet Explorer). See #39 .
The Windows service doesn't start	Check log-file in the service's hub.conf, and Windows Event Viewer > Windows Logs > System for Service Control Manager errors.

Decoding BACnet/SC traffic in Wireshark

BACnet/SC is TLS 1.3, so a capture shows only encrypted records. To see the BACnet/SC messages inside, have the hub write each connection's TLS session secrets to a key log file, and give that file to Wireshark:

```
BACnetExampleBSCHUB --sc-keylog-file sc-keys.log
```

1. Start the capture (for example tcp port 4443; on Windows, capture traffic to the same computer with Npcap's loopback adapter), then start the hub with --sc-keylog-file. Only connections made while the option is on can be decoded, because Wireshark needs each TLS handshake.
2. In Wireshark: **Edit > Preferences > Protocols > TLS > (Pre)-Master-Secret log filename**, and select sc-keys.log. With tshark: tshark -r capture.pcapng -o tls.keylog_file:sc-keys.log.
3. The records then decode as WebSocket, and the BACnet/SC frames as **BSCVLC** (Connect-Request, Connect-Accept, Encapsulated-NPDU, ...). Filter with bscv1c. If you run BACnet/SC on a port Wireshark doesn't recognise, use **Decode As > TCP port > TLS**.

The file covers both BACnet/SC roles: devices connecting to the hub, and the hub's own connection to another hub (--sc-hub-uri). It doesn't cover the HTTPS status page. The hub appends to the file and flushes every line straight away, so you can decode a capture while the hub is still running.

Anyone with this file and a capture can read that traffic. Use the option for debugging only, then delete the file. It never holds a private key. It can be set on the command line only (there is no config-file key), so a service can't be left logging by a forgotten line in its config. While it is on, the hub logs a warning at start-up and whenever BACnet/SC starts listening, /health reports "tls_keylog": true, and the status page shows a banner.

11. The BACnet device

Device 389022	"Chipkin Example B-SCHUB"	Vendor 389 (Chipkin Automation Systems)
└─ Analog Input 1	"Bronze"	example sensor value, degrees C
└─ Network Port 1	"BACnet IP"	BACnet/IP, UDP 47808 (absent with --bacnet-ip off)
└─ Network Port 2	"BACnet SC"	BACnet/SC hub function (and optional connector)

└─ File 1	"Operational Certificate"	the hub's certificate	(writable)
└─ File 2	"Certificate Signing Request"	CSR for the hub's key	(read-only)
└─ File 3	"Issuer Certificate Slot 1"	trusted CA certificate	(writable)
└─ File 4	"Issuer Certificate Slot 2"	second trusted CA certificate	(writable)

Every object has a **Description** saying what it is for. The Device's Description links back to the project.

The device instance defaults to 389022 (`--deviceId`), and the Device's name to "Chipkin Example B-SCHUB" (`--device-name`). Both must be unique on the BACnet internetwork. Analog Input 1 is example data: the up/down arrow keys change it while the hub runs.

Each Network Port reports `Network_Number 0` with `Network_Number_Quality` **unknown** unless `--ip-network-number / --sc-network-number` sets it; this device isn't a router. The private key is never served by any object. The [PICS](#) lists every property.

12. Version, licensing and support

- **Version:** this manual describes version 1.5.0. What changed in each release is in [CHANGELOG.md](#); what a version number promises is in [SUPPORT.md](#).
- **Licensing:** the hub's own source code is public domain (CC0-1.0). It is built on the **CAS BACnet Stack**, a commercial Chipkin product, and on OpenSSL (Apache-2.0), libwebsockets (MIT), libuv (MIT), zlib (Zlib) and, on Windows, pthreads4w (Apache-2.0) - see [THIRD-PARTY-NOTICES.md](#). Each release's `sbom-*.cdx.json` lists the exact versions.
- **Support:** questions and bug reports go to [GitHub issues](#). For commercial support, a production hub or a CAS BACnet Stack licence, contact support@chipkin.com. Security problems: [SECURITY.md](#).
- **Related documents:** the [fact sheet](#), the [PICS](#), the [production certificate guide](#) and the [BTL test plan](#).